The Incompleteness of Observation

Why Physics' Biggest Contradiction Might Not Be a Contradiction at All

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0.1 The Worst Prediction in Physics

Physics has two spectacularly successful theories. Quantum mechanics describes the behavior of atoms, particles, and light. General relativity describes gravity, space, and time. Each has been confirmed to extraordinary precision. They have never disagreed with any experiment.

They disagree with each other.

Ask quantum mechanics how much energy empty space contains and it gives you a staggering number: roughly 10^{113} joules per cubic meter. Ask general relativity the same question — read the answer off the expansion rate of the universe — and you get about 6×10^{-10} joules per cubic meter. The ratio is 10^{122}. For scale, the number of atoms in the observable universe is about 10^{80}. This is not a close call.

For decades, the assumption has been that something is deeply broken — that one or both calculations contain an error, and that finding the mistake will point the way to a unified theory of everything.

This paper argues the opposite. Neither calculation is wrong. They disagree because they *must*. In fact, that massive 10^{122} discrepancy isn't a failure at all. It is the strongest piece of existing evidence we have that quantum mechanics is not the fundamental bedrock of reality, but an emergent description forced upon us by our limited vantage point.

The argument is built from a chain of mathematical proofs, each feeding into the next. This document explains what the paper claims, walks through the logic of every major proof, and shows how they connect.

0.2 The Starting Point

Every mathematical proof starts from assumptions, and the paper is explicit about its four. None of them mention quantum mechanics. None of them mention general relativity. They are:

1. **Deterministic dynamics.** The universe evolves according to definite rules. Given the complete state at one time, the state at any other time is uniquely determined. This is just classical physics — Hamilton's equations on a phase space.
2. **Finiteness.** The part of the universe the observer can describe has a finite number of distinguishable configurations. This is motivated by a purely geometric fact: a finite-area boundary can only couple to finitely many modes across it. The paper treats the discreteness scale ϵ as a free parameter and derives its value later.
3. **Causal partition.** The universe splits into what the observer can access (the visible sector) and what they can't (the hidden sector). The two sectors interact, but the observer can only measure one of them.
4. **Classical probability.** The observer reasons using ordinary probability theory — expectation values, the law of large numbers, the central limit theorem. Nothing exotic.

That's it. The claim is that quantum mechanics — the Schrödinger equation, the Born rule, superposition, entanglement, Bell inequality violations — follows from these four premises alone, given the right conditions on the hidden sector. Specifically, the hidden sector must be:

- **Coupled** (C1) — it dynamically interacts with the visible sector, not just passively coexists with it.
- **Slow** (C2) — it retains correlations longer than the timescale of any experiment, so its memory is still there between measurements.

- **Vast** (C3) — it has enough capacity that its memory never fills up or gets overwritten.

The axioms set the stage. The conditions determine what kind of show plays on it. The next section explains why the cosmological horizon satisfies all three.

0.3 The Observer's Blind Spot

Light travels at a finite speed, and the universe has a finite age. Put those together and every observer has a horizon — a boundary beyond which no signal has had time to arrive. Everything beyond that boundary is ordinary physics: fields, particles, radiation. But it is structurally inaccessible. Not because our telescopes aren't good enough, but because the geometry of spacetime forbids it. No technology that obeys the speed of light can reach past it.

This means every observer in the universe is in the same epistemic situation: there are degrees of freedom — a vast number of them — that influence what you measure but that you can never track. When you write down the laws of physics for the things you *can* see, you're forced to average over everything you can't. You have to "trace out" the hidden sector.

Here's what that looks like concretely. The total system — visible plus hidden — is deterministic. If you knew the complete state, you could predict the future exactly. But you don't know the hidden part. You know the visible state is x, but there are many possible hidden states compatible with x, and each one sends x to a different visible future. Hidden state h_1 might send the particle left; hidden state h_2 might send it right. Since you can't tell which hidden state you're in, the best you can do is assign probabilities: average over all the possible hidden states, weighted by how likely each one is. The result is a set of *transition probabilities* — the chance that visible state x at time t_1 becomes visible state y at time t_2. You've gone from a deterministic system you can't fully see to a probabilistic one you can. That's a stochastic process, and it's the only description available to any observer who can't access the hidden sector.

The standard expectation is that this should produce something boring — classical, memoryless noise. And it would, if the hidden sector were fast and forgettable, like air molecules bouncing off a grain of pollen. Each kick is independent of the last. Physicists call this *Markovian* behavior.

But the hidden sector beyond the cosmological horizon is not like that. It differs in three specific ways, and the paper proves that their conjunction changes everything.

It's coupled. The horizon is not a static wall. Stress-energy conservation enforces continuous dynamical correlations across it. Matter crosses the horizon, and the horizon area adjusts in response to interior energy density. Information flows in both directions. (Condition C1.)

It's slow. The hidden sector's correlation time is set by the Hubble timescale — roughly 10^{17} seconds, the age of the universe. Any laboratory experiment operates on timescales of 10^{-15} seconds or shorter. The ratio is 10^{-32}. The hidden sector cannot "reset" between your measurements. Every correlation it picks up from one experiment is still there when the next one begins. This is the *opposite* of the standard Markovian regime, where the environment decorrelates fast. Here, it never decorrelates at all. (Condition C2.)

It's vast. The hidden sector has roughly 10^{122} independent degrees of freedom — the Bekenstein-Hawking entropy of the cosmological horizon. No experiment you could ever perform would appreciably disturb its state. Its memory never saturates. (Condition C3.)

A fast environment with vast capacity would wash out correlations (Markovian noise). A slow environment with limited capacity would eventually fill up and stop recording. Only an environment that is simultaneously coupled, slow, and vast sustains the kind of persistent, non-

decomposable correlations that the paper calls *P-indivisibility* — a technical term meaning the system's transition probabilities at different times cannot be broken into independent steps.

0.4 From Indivisibility to Quantum Mechanics

This is the key link, and it can be established by two independent mathematical routes.

The first route uses a result proved by Jacob Barandes in 2023. Barandes showed that there is an exact equivalence between two things that seem completely different. On one side: any stochastic process whose transition probabilities are P-indivisible — that is, any process with the kind of persistent, undecomposable memory described above. On the other side: a quantum system evolving unitarily under the Schrödinger equation.

The second route uses Stinespring's dilation theorem, a foundational result in functional analysis from 1955. Because the total dynamics is a deterministic bijection on a finite set, it defines a unitary operator on a Hilbert space. Tracing out the hidden sector with the Liouville measure produces a quantum channel — a completely positive, trace-preserving map — whose measurement probabilities are identical to the classical transition probabilities. The Stinespring construction requires nothing beyond textbook mathematics.

The equivalence is not approximate. It is not an analogy. It is a mathematical identity. Every P-indivisible stochastic process *is* a quantum system, and every quantum system *is* a P-indivisible stochastic process. The Schrödinger equation, the Born rule, interference, entanglement, and superposition all come out automatically.

So the argument chains together like this:

1. The cosmological horizon partitions the universe into what you can see and what you can't.
2. Tracing out the hidden sector produces a stochastic process — you're forced to assign probabilities because you can't track the hidden degrees of freedom.
3. The three conditions (coupled, slow, vast) guarantee that this stochastic process is P-indivisible.
4. By either Barandes' correspondence or the Stinespring construction, P-indivisible stochastic dynamics is mathematically equivalent to quantum mechanics.

Quantum mechanics is not a fundamental law of nature. It is what embedded observation looks like — and the paper proves the equivalence runs both ways. These three conditions produce quantum mechanics, and any quantum system, traced back to its deterministic roots, requires exactly these conditions. Quantum mechanics and embedded observation are the same thing.

0.5 Inside the Proofs

The argument above sounds clean in outline. But each link is a rigorous mathematical proof, and the proofs are where the real persuasion lives. The rest of this document walks through each one in detail — what it claims, why it's true, and what it feeds into next.

0.5.1 Proof 1: The P-Indivisibility Theorem (§2.3)

What it claims. If a deterministic system is split into a visible and hidden sector, and these sectors are genuinely coupled, then the visible sector's behavior *cannot* be a simple memoryless random process. It must exhibit P-indivisibility — a specific kind of built-in memory.

What "P-indivisible" means. Imagine watching someone roll dice. If the dice are fair and independent, each roll has no memory of the last — knowing what happened on roll 5 tells you nothing about roll 6. This is a "Markov process," or equivalently, a "P-divisible" one. Its key property is that distinguishable things always stay distinguishable or become less so over time. Two different starting points gradually blur together, and they *never* un-blur.

P-indivisibility means the blurring *reverses*. Two states that looked identical at some intermediate time later become distinguishable again. The system "remembers" distinctions that appeared to be erased.

The three-step proof. The proof is surprisingly short. It uses only two ingredients: the dynamics are deterministic and reversible, and the visible and hidden sectors are genuinely coupled.

Step 1 — Everything comes back. The total system is deterministic and operates on a finite number of states. A reversible shuffling of a finite deck of cards must eventually return to the original order — there are only so many arrangements. After some number of steps N, every state is back where it started. Distinguishability is fully restored.

Step 2 — But things blur in between. Because the visible and hidden sectors are coupled, the same visible state can lead to different visible outcomes depending on which hidden state happens to be in play. Since the observer doesn't know the hidden state, they see probabilistic mixing. Distinguishability has decreased.

Step 3 — The contradiction with Markov behavior. After one step, distinguishability went down. After N steps, it's fully restored. That means it went down and then came back up. But in any P-divisible (memoryless) process, distinguishability can only decrease or stay the same. The fact that it increases proves the process is P-indivisible.

0.5.2 Seeing It Work: The Coin-and-Dice Model (§2.4)

The paper builds a concrete toy model to make the mechanism tangible.

The setup. The visible sector is a single coin: Heads (0) or Tails (1). The hidden sector is a six-sided die (values 1 through 6). The total system has 12 states. The dynamics are a deterministic rule that repeats after two steps. Two of the six die values (1 and 2) cause the coin to flip; the other four leave the coin alone but shuffle the die.

After one step: The observer doesn't know the die. Since 2 out of 6 hidden states flip the coin and 4 don't, the coin has a 1/3 chance of flipping and a 2/3 chance of staying. There's genuine randomness — from the observer's perspective.

What a memoryless process would predict after two steps: Continued blurring. Starting at Heads, there'd be roughly a 44% chance of being at Tails. Things should keep mixing.

What actually happens after two steps: Every state returns to its starting point. The coin is back where it started with 100% certainty. Complete un-mixing.

The smoking gun: To bridge the gap between the one-step and two-step results, you'd need a valid intermediate propagator. The paper computes it and finds it contains *negative entries* (-1 in the off-diagonals). Probabilities can't be negative. No valid probability matrix exists. The process is P-indivisible.

The mechanism: The die works as a memory register. At step 1, die values 1 and 2 flip the coin while preserving the die — so after step 1, a coin at Tails with die value 1 carries an implicit record: "I was Heads." At step 2, the same rule reads that record and flips the coin back. The hidden sector stored which transitions happened and played them back.

0.5.3 Why conditions C2 and C3 matter physically

The theorem above needs only coupling (C1) and finiteness. So why does the paper insist on slow memory (C2) and vast capacity (C3)?

Because P-indivisibility without C2 and C3 might only show up at absurd timescales or might self-destruct. C2 ensures the memory persists on timescales accessible to actual experiments, not just at cosmic recurrence times. C3 ensures the hidden sector never runs out of room to store information — if it saturates, later imprints overwrite earlier ones, and the process becomes effectively memoryless. Together, C2 and C3 guarantee that P-indivisibility is strong, persistent, and observationally relevant.

0.5.4 Proof 2: The Stochastic-Quantum Correspondence (§3.1 and Appendix A)

What it claims. Any P-indivisible stochastic process is mathematically identical to a quantum system evolving unitarily. There exists a Hilbert space and a unitary operator $U(t)$ such that the transition probabilities of the stochastic process are *exactly* the Born-rule probabilities: $T_{ij}(t) = |U_{ij}(t)|^2$.

Two independent routes to the same conclusion. The primary route uses Barandes' stochastic-quantum correspondence (2023–2025), which establishes the equivalence for any P-indivisible process. The secondary route, given in Appendix A, uses Stinespring's dilation theorem (1955): a deterministic bijection on a finite product space defines a permutation unitary; tracing out the hidden sector with the Liouville measure produces a completely positive quantum channel whose diagonal elements recover the classical transition probabilities exactly. This second route requires only textbook results from the 1950s–2000s. Either route alone suffices; together they ensure the bridge rests on no single recent result.

The intuition. P-indivisibility means transition probabilities can't be factored through intermediate times — try it and you get "negative probabilities." In standard probability theory, this is nonsensical. But in quantum mechanics, it's *exactly what happens*: probability amplitudes (which can be negative or complex) combine to produce interference patterns that don't factorize classically. What Barandes proved is that these are not analogous phenomena — they are the same mathematical object, written in different notation.

What emerges for free. Three pillars of quantum mechanics come out of this correspondence without being assumed:

- *The Schrödinger equation.* The process evolves continuously, so the unitary operator is differentiable. The Schrödinger equation is the unique time-evolution law compatible with unitarity and continuity.
- *The Born rule.* In standard quantum mechanics, "probability equals amplitude squared" is an additional postulate. Here it's derived: $T_{ij} = |U_{ij}|^2$ is the definition of how the stochastic process maps to the quantum description.
- *Bell inequality violations.* Since the transition matrices for composite systems don't factorize, entangled systems naturally produce correlations that violate Bell inequalities, up to exactly Tsirelson's bound — the maximum quantum violation.

What does *not* emerge yet. The value of Planck's constant $\hbar$. The correspondence says "a quantum description exists" but is silent about the dimensionful scale. Fixing $\hbar$ requires the physical content of the partition — the job of §5.

0.5.5 Proof 3: The Characterization Theorem (§3.3)

What it claims. The paper's main theorem says: *if* you have a deterministic system with coupling, slow memory, and sufficient capacity, *then* the observer sees quantum mechanics. The characterization theorem says the reverse: *if* a process is quantum mechanical, it *must* arise from embedded observation under exactly those three conditions. The equivalence is exact: QM $\iff$ embedded observation under C1–C3.

Why each condition is independently necessary:

C1 (coupling). If the hidden sector doesn't affect the visible sector, each visible state maps deterministically to exactly one other. That's a permutation — perfectly decomposable, no memory, P-divisible. No coupling $\to$ no quantum mechanics.

C2 (slow memory). If the hidden sector scrambles itself between interactions — like re-rolling the die before every coin flip — each step becomes independent. That's a Markov chain. The paper formalizes this via the ergodic theorem: fast mixing drives the hidden sector's state to uniformity, erasing all correlations. No memory $\to$ no quantum mechanics.

C3 (sufficient capacity). The hidden sector is the only place memory can be stored (the visible sector is known to the observer). The total non-Markovian memory is bounded by the hidden sector's information capacity: $I(\text{past}; \text{future} \mid \text{present}) \leq \log_2(\text{hidden sector size})$. If the hidden sector is too small, it runs out of room. Earlier records get overwritten. The process becomes effectively Markov. Insufficient capacity $\to$ no quantum mechanics.

The full equivalence. The three pieces snap together by transitivity:

- Barandes' correspondence / Stinespring construction: QM $\iff$ P-indivisibility
- Sufficiency (§2.3): embedded observation (C1–C3) $\implies$ P-indivisibility
- Necessity (the three results above): P-indivisibility $\implies$ embedded observation (C1–C3)

Therefore: **QM $\iff$ embedded observation under C1–C3.** There's no way to get quantum mechanics from a deterministic substrate except through these conditions, and no way to have these conditions without getting quantum mechanics.

0.6 Where Planck's Constant Comes From

The paper proves a further result called *partition-relativity*: the emergent quantum description is completely and uniquely determined by the partition — the boundary between what the observer can and cannot see. Two observers sharing the same cosmological horizon get exactly the same quantum mechanics. Different partition, different quantum mechanics (or, if the conditions aren't met, no quantum mechanics at all).

This means $\hbar$ — Planck's constant, the fundamental scale of quantum mechanics — cannot be a free parameter. It must be fixed by the geometry of the boundary.

0.6.1 Proof 4: The Volume-Independence Lemma and the $\hbar$ Derivation (§5.2)

The problem. The stochastic-quantum correspondence tells us quantum mechanics emerges, but not the value of Planck's constant. $\hbar$ has units of energy $\times$ time, and the correspondence operates entirely in dimensionless probabilities. Something external must set the scale. But *which* properties of the boundary determine it? The hidden sector is vast (the entire universe beyond the horizon). Does $\hbar$ depend on the total amount of stuff out there?

The Volume-Independence Lemma: Only the boundary matters. The answer is no. The formal proof has three parts. First, spatial locality of the classical Hamiltonian means the visible sector couples only to boundary modes — deep modes must propagate through the boundary layer before influencing anything observable, and the coupling chain is $V \leftrightarrow B \leftrightarrow D$ with no direct V-D term. Second, condition C2 means the deep sector is effectively frozen on experimental timescales: it evolves by $\mathcal{O}(t/\tau_B) \sim 10^{-32}$ per laboratory measurement. Third, because the deep modes are frozen, the sum over them in the transition-probability formula contributes identical terms that cancel in normalized probabilities. The transition probabilities depend only on the boundary modes.

The four-step derivation. With volume excluded:

Step 1 (Uniqueness). Partition-relativity guarantees the quantum description is unique for a given partition. $\hbar$ is determined, not free.

Step 2 (Structural, not volumetric). The volume-independence lemma excludes any dependence on H (which sets the horizon's size). If $\hbar$ depended on H, observers at different cosmic epochs would have different quantum mechanics. The only candidates are c, G, and ϵ.

Step 3 (Dimensional analysis). There's only one way to combine c, G, and ϵ to get units of action: $c^3\epsilon^2/G$, times a dimensionless prefactor β.

Step 4 (Thermal self-consistency). The horizon has a classical temperature derivable from GR alone, with no $\hbar$. Independently, the emergent quantum theory predicts a Gibbons-Hawking temperature that depends on $\hbar$. Requiring these to agree gives one equation in one unknown:

$$\hbar = \frac{c^3\epsilon^2}{4G}$$

The prefactor is $\beta = 1/4$. No free parameters remain. The discreteness scale pins to $\epsilon = 2\,l_p$ — twice the Planck length. The Bekenstein-Hawking entropy formula $S = A/(4\,l_p^2)$ — including its famous factor of 1/4 — falls out as a direct consequence.

Why this isn't circular. The Gibbons-Hawking temperature is a result of quantum field theory — the very theory being derived. But the logic is sequential: Part I proves quantum mechanics emerges with *some* $\hbar \to$ the emergent QM predicts $T_{\mathrm{GH}}(\hbar) \to$ the classical temperature T_{cl} was calculated independently $\to$ matching them fixes $\hbar$. This is a "gap equation," the same structure used routinely in condensed matter physics.

0.6.2 Proof 5: The D-Gauge Completeness Theorem (§5.3)

The problem. Barandes' correspondence maps probabilities to quantum amplitudes, but the map has built-in ambiguity. Transition probabilities constrain only $|U_{ij}|^2$ — the squared magnitudes. The quantum *phases* (the angles in the complex plane) are not determined by probabilities alone. This is the "Schur-Hadamard gauge freedom." If the phases are ambiguous, the emergent quantum description isn't unique.

The resolution: Continuous time locks the phases. The paper's setup provides the full continuous-time evolution of transition probabilities, $T_{ij}(t)$ for all times t. The theorem shows this is enormously more constraining than single-time data.

The unitary $U(t) = e^{-iHt}$ can be expanded in the energy eigenbasis. The transition probabilities become Fourier series in time, with frequencies given by energy differences. If the energy

spectrum is non-degenerate (a condition holding for almost all systems), the Fourier frequencies are distinct and the coefficients are uniquely determined. Working through the algebra uniquely fixes the moduli and constrains the phases to a single remaining freedom: a global rephasing of the basis states (like choosing where to set the zero on your clock). This changes nothing observable — all transition probabilities, energy spacings, and interference patterns are invariant.

The dimensional obstruction. Even with all phases locked, the probability data cannot determine the overall dimensionful scale. $\hbar$ enters only when converting the dimensionless unitary into a physical Hamiltonian. No amount of dimensionless data can produce a dimensionful constant. This is why Step 4 of the $\hbar$ derivation (thermal self-consistency) is not just a convenient check but the *mathematically obligatory* step: it's the only place where dimensionful physical input enters the framework.

0.7 The Two Levels

Now the cosmological constant problem dissolves. But first, a crucial link: how does finite-dimensional quantum mechanics become quantum *field* theory — with independent modes, each carrying zero-point energy?

The answer is spatial locality. The classical substratum has it — neighboring cells interact, distant cells don't. The paper proves that the emergent quantum description inherits it. The argument is direct: the Barandes correspondence maps each classical configuration $(x_1, x_2, \ldots, x_N)$ to a quantum basis state $|x_1, x_2, \ldots, x_N\rangle$. That's already a tensor product. The only question is whether the emergent Hamiltonian respects the structure. And it must, because if two configurations differ only at sites that aren't neighbors, the classical dynamics can't connect them in an infinitesimal time step — so the transition probability between them is zero — so the corresponding quantum Hamiltonian matrix element is zero. The emergent Hamiltonian couples only neighboring cells, exactly like the classical one. You get a lattice quantum field theory with a built-in ultraviolet cutoff at $\epsilon = 2\,l_p$.

This is what makes the CC dissolution work. The QFT has independent modes. Each mode gets a zero-point energy. The sum diverges. And the "worst prediction in physics" follows — but only within the emergent description.

The critical insight is that general relativity and quantum field theory are not competing answers to the same question. They are answers to *different questions*, asked at different levels of the same reality.

Level 1: The classical substratum. Spacetime geometry is part of the fundamental layer. The metric tensor evolves via Einstein's field equations. The stress-energy tensor that sources gravity is the classical stress-energy of the total microstate. At this level, the vacuum energy density sits at the critical scale: $\rho \sim H^2/G \sim 10^{-9}$ J/m^3. No zero-point energy. No discrepancy.

Level 2: The emergent quantum description. For an embedded observer tracing out the hidden sector, the mandatory quantum description assigns a zero-point energy of $\frac{1}{2}\hbar\omega$ per mode. Sum to the Planck cutoff and you get $\rho_{\mathrm{QFT}} \sim 10^{113}$ J/m^3. This number is real *within the quantum description* — it reflects the magnitude of the trace-out noise — but it is not a source term in Einstein's equations, because those equations operate at the classical level, which is logically prior to the quantum description.

This ordering is not a choice but is forced by three independent requirements. First, the partition must be definite — not in superposition — for the trace-out to be well-defined; a partition in superposition would yield an incoherent mixture of inequivalent quantum theories. Second, the

partition is defined by the causal structure, which is determined by null geodesics of the metric; if the metric were derived from QM, the derivation would be circular (QM → metric → causal structure → partition → QM). Third, $\hbar$ is determined by the boundary geometry; if the geometry were itself quantum-mechanical, $\hbar$ would depend on a quantum state, contradicting its observed universality.

The 10^{122} ratio between the two answers is not a discrepancy. It equals S_{dS} — the Bekenstein-Hawking entropy of the cosmological horizon — which is the number of hidden-sector degrees of freedom the trace-out compresses into the emergent quantum state. The "worst prediction in physics" is the information compression ratio of the observer's blind spot. A category error, not a fine-tuning failure.

This is not a prediction awaiting future data. The observed vacuum energy has been measured since 1998, and it sits exactly at the classical geometric scale — the value the framework expects. Meanwhile, every attempt to search for a cancellation mechanism within quantum-first frameworks have found nothing. The framework explains why: there is nothing to cancel.

This track record is itself evidence for the ordering. The question of whether geometry is prior to quantum mechanics or quantum mechanics is prior to geometry is usually treated as a philosophical preference. But it has an empirical signature sitting in plain sight: one ordering produces the worst prediction in physics and has no solution; the other predicts the observed value and explains the discrepancy as a derived quantity. That doesn't prove the ordering is correct — the gravitational wave echo prediction provides a more direct test — but it's existing evidence, not a future hope.

0.8 What Quantum Weirdness Looks Like From Here

If quantum mechanics is an emergent description forced on embedded observers, the standard quantum puzzles acquire straightforward readings.

The double-slit experiment. The particle goes through one slit. The interference pattern arises because the transition probabilities are computed by marginalizing over the hidden sector, and the hidden sector includes the field configuration near both slits. Opening or closing the second slit changes the boundary conditions, which changes the marginalization, which changes the pattern. The "wave-like" behavior is the hidden sector's influence shifting when the geometry changes.

Entanglement. Two particles prepared together share a joint transition matrix inherited from the trace-out. The correlations are encoded in the structure of the dynamics itself, not in a hidden variable you could integrate over. This is why Bell inequality violations occur: the standard factorization assumption fails for indivisible processes. The framework reproduces quantum correlations exactly up to Tsirelson's bound, without faster-than-light signaling and without superdeterminism.

The measurement problem. Measurement produces definite outcomes through the indivisible dynamics. When Wigner can't access his Friend's lab, he traces out its internal degrees of freedom and assigns a superposition. The superposition reflects Wigner's epistemic situation — what *he* can infer — not the Friend's physical state. Branching in the Many-Worlds sense is a feature of the compressed description, not of the underlying reality.

0.9 Predictions

The framework is not just a reinterpretation. It makes specific, falsifiable predictions — and, crucially, it makes predictions in domains it was not designed to address.

Dark energy evolution. Because the hidden sector's dimensionality changes as the Hubble parameter H evolves (the horizon area is $A = 4\pi c^2/H^2$), the emergent vacuum energy inherits a dependence on H. The predicted form matches the Running Vacuum Model: $\Lambda_{\text{eff}} = \Lambda_0 + \nu H^2$. The coefficient ν is computed from the spectral structure of the horizon: the trace-out compression noise is distributed over $\mathcal{N} = \ln(c/\epsilon H)$ spectral decades, giving a conformal benchmark of $\nu \approx 2.45 \times 10^{-3}$ with a systematic range of $\sim 1\text{–}5 \times 10^{-3}$ (the dominant uncertainty is the spectral assumption, not the cosmological parameters). DESI's 2024–2025 data releases report evidence for evolving dark energy at 2.8σ–4.2σ, with phenomenological fits consistently finding $\nu \sim 1\text{–}4 \times 10^{-3}$ — the range in which the framework's prediction sits.

Gravitational wave echoes. At a proper distance of about one discreteness scale ($\epsilon = 2\,l_p$) outside a black hole horizon, an infalling mode's wavelength hits the discreteness floor and must scatter back. For a 30 solar-mass remnant, the predicted echo delay is on the order of 50 ms.

The dark sector as corroboration. The trace-out that produces quantum mechanics has a gravitational consequence that the paper did not set out to find. The boundary entropy — the S_{dS} modes traced out to produce the quantum description — has thermal energy that, distributed over the Hubble volume, equals the critical density exactly. A crucial subtlety: this thermal energy is computed entirely from pre-trace-out (classical) quantities — the number of boundary modes, the classical horizon temperature, and the Hubble volume — with no reference to $\hbar$ or the emergent quantum description. This is what distinguishes the boundary entropy's gravitational contribution from the QFT zero-point energy ($\rho_{\text{QFT}} \sim 10^{113}$ J/m^3), which exists only *after* the trace-out and is an artifact of the emergent description. The framework denies that the zero-point energy gravitates (this is the CC dissolution of Part II); the boundary entropy's classical thermal energy *does* gravitate, at the scale $\rho_{\text{crit}} \sim 10^{-9}$ J/m^3.

The paper proves that this entropy has no operator in the emergent QFT. The baryonic sector — what QFT can account for — is ~5% of ρ_{crit}. The remaining ~95% is the boundary entropy: gravitationally active, invisible to the emergent description, and persistent through P-indivisibility (condition C2). This matches the observed composition of the universe, in which ~95% of the gravitational content has no source in particle physics. The uniform component corresponds to dark energy (handled by Part II's CC dissolution); the structured component (dark matter) is a candidate consequence of matter-induced entropy redistribution, though the redistribution profile requires further work. That axioms designed to derive quantum mechanics also account for the dark sector's total budget is independent corroboration that observational incompleteness is capturing real structure.

The conjunction is distinctive: dark energy evolution in RVM form, gravitational wave echoes, and a natural account of the dark sector's total gravitational budget — all from four axioms about embedded observation. No competing framework produces all three. The parallel with the cosmological constant dissolution is exact. Part II identifies the 10^{122} discrepancy as the *information compression ratio* of the trace-out — how much the emergent description distorts the vacuum energy. Part III identifies the ~95% dark sector as the *gravitational occlusion fraction* — how much of the gravitational universe the trace-out hides from the emergent description. Together, they account for the two largest anomalies in modern cosmology as two aspects of a single phenomenon: the cost of observing the universe from within.

0.10 Philosophical Lineage

The paper is a physics paper, but its core claims — that observers face irreducible limits, that two irreconcilable descriptions can both be correct, that incompleteness is a structural feature rather than a deficiency — sit at the intersection of some of the oldest debates in philosophy. A systematic mapping against the major traditions reveals a striking pattern: broad support for most of the framework, and near-universal resistance to one specific thesis.

0.10.1 The seven claims

The framework rests on seven implicit philosophical commitments:

1. **Embedded observers face irreducible limits.** No observer inside a system can access the complete state.
2. **QM and GR are both correct** within their domains.
3. **The hidden sector is permanently inaccessible** — not due to technological limitations, but structural ones.
4. **The underlying reality is local and definite.** Indeterminacy belongs to the observer's description, not to the world.
5. **Incompleteness is structural, not deficient.** The limitation arises because the observer is made of the same elements as the universe it is trying to describe — a physical form of self-reference. This is analogous to Gödel's incompleteness theorem, not to ignorance that better instruments could cure.
6. **The description is observer-relative.** Different partitions yield different emergent physics.
7. **The two descriptions are irreconcilable** — not because one is wrong, but because they are complementary projections of a single reality that no embedded observer can access directly.

Claims 1, 5, 6, and 7 enjoy broad philosophical support across nearly every tradition examined. Claim 4 — that the underlying reality is definite — is the paper's most philosophically isolated thesis.

0.10.2 The Gödel connection

The analogy between this framework and Gödel's incompleteness theorem is not merely metaphorical. Gödel proved that a formal system rich enough to encode arithmetic cannot prove all true statements about itself from within — the limitation arises because the system is self-referential, capable of constructing sentences that refer to its own provability. The observer in this framework faces a structurally parallel situation: the reason a cosmological horizon exists at all is that the observer is a physical subsystem made of the same fields, obeying the same speed-of-light constraint, as the universe it is trying to describe. An observer not made of the universe's own elements would not face a horizon and would not be forced into a quantum description. The incompleteness is a consequence of self-inclusion.

The connection can be made precise through Wolpert's limits of inference [19], which the paper cites. Wolpert proved, using diagonal self-referential arguments directly descended from Gödel's, that any inference device embedded in the universe it is trying to predict faces fundamental limits — not because of noise or finite resources, but because complete self-prediction is logically impossible. The present framework provides the concrete physical mechanism by which Wolpert's logical limitation manifests: the causal partition enforces the trace-out, the trace-out produces P-indivisibility, and P-indivisibility is quantum mechanics. The 10^{122} compression ratio — the Bekenstein-Hawking entropy of the cosmological horizon — is the quantitative measure of how much information self-inclusion forces the observer to lose.

The key difference from Gödel is in the *form* of self-reference. Gödel's is syntactic: the system encodes a sentence that says "I am not provable." The framework's is physical: the observer is made of the described, so complete description would require the observer to fully encode its own state plus everything causally connected to it, which the causal structure forbids. Wolpert sits between the two, using Gödelian logic applied to physical systems. Together they form a chain: Gödel (formal systems cannot completely describe themselves) → Wolpert (physical inference devices cannot completely predict the systems they inhabit) → this framework (the specific mechanism is the causal partition, and the specific cost is quantum mechanics).

Hofstadter's *Gödel, Escher, Bach* argues that self-referential systems produce genuinely emergent higher levels — "strange loops" where a system's hierarchical levels fold back on themselves, generating properties invisible at the lower level. This framework agrees: quantum mechanics is the real, irreducible description available to any embedded observer, just as Hofstadter's "I" is real even though it emerges from neurons. The strange loop is intact — the observer, made of the universe's own elements, generates through self-inclusion an emergent description that governs everything the observer can access. The disagreement with Hofstadter is narrow but consequential: he argues that once the emergent level is established, the substrate beneath it is explanatorily inert. The cosmological constant problem suggests otherwise. The 10^{122} discrepancy exists precisely because the emergent quantum description and the classical substrate assign different vacuum energies, and only the substrate's value matches observation. The loop has a floor, and the floor matters.

0.10.3 Closest ancestor: Nicholas of Cusa

Of all thinkers surveyed, the fifteenth-century cardinal Nicholas of Cusa provides the most precise structural alignment. His *docta ignorantia* (learned ignorance) is essentially Claim 5: the highest knowledge is knowing what we cannot know, and this is an intellectual achievement, not a failure. His *coincidentia oppositorum* (coincidence of opposites) maps onto Claim 7 with remarkable precision: contradictions irreconcilable in the finite realm dissolve in infinity — the infinitely large circle's circumference becomes a straight line, the infinite polygon becomes a circle. QM and GR, irreconcilable within any finite observational framework, would be coincident in the infinite ground that generates both.

Most strikingly, Cusa's "wall of Paradise" from *De Visione Dei* maps onto Claim 3: an insurmountable boundary beyond which finite intellect cannot pass. Scholars like Emmanuel Falque interpret reaching this wall not as escaping through it but as *inhabiting the boundary* — profoundly compatible with a framework where understanding the limit is itself the deepest available insight.

0.10.4 Deepest ontological parallel: Spinoza

Spinoza's attribute theory is arguably the single closest ontological parallel. One substance (God/Nature) expresses itself through infinite attributes, of which humans know only two: Thought and Extension. Jonathan Bennett's "barrier doctrine" captures the key feature: each attribute must be conceived through itself — no explanatory flow crosses between them. This is precisely Claim 7: two complete, correct descriptions that are structurally irreconcilable, yet both describe the same underlying reality. Spinoza's substance is fully determinate, aligning with Claim 4. His parallelism doctrine — the order of ideas is the same as the order of things — means the two descriptions track the same structure through incommensurable vocabularies.

The divergence is epistemological: Spinoza believes reason achieves adequate knowledge of reality through *scientia intuitiva* (intuitive knowledge). The paper's permanent limits directly contradict this ambition.

0.10.5 The recurring fault line: Claim 4

Across every tradition examined, a striking pattern emerges. The claim that underlying reality is "local and definite" faces resistance from virtually every direction:

Kant prohibits positive characterization of the noumenal realm as dogmatic metaphysics — you can know *that* things-in-themselves exist but never *what* they are. **Hegel** diagnoses a performative contradiction: to posit a hidden sector and characterize it as containing standard physics is already to have crossed the boundary you claim is uncrossable. **Nietzsche** attacks it as residual Platonism — having correctly shown that all observation is perspectival, the paper reinstates the very "true world" his career was dedicated to destroying. **Wittgenstein** rejects it as nonsensical: asserting what lies beyond the limits of the sayable transgresses exactly the limits the paper identifies.

Nagarjuna identifies it as *svabhava*-reification — attributing inherent existence to what Buddhist emptiness (*sunyata*) says lacks it. **Daoism** warns against naming the unnameable: the Dao that can be spoken is not the eternal Dao, and calling reality "definite" is itself an act of conceptual carving. **Whitehead** insists reality is fundamentally processual and creative, involving genuine indeterminacy — removing that indeterminacy robs the cosmos of its creative character.

Advaita Vedanta comes closest to full support — Brahman is indeed a definite underlying reality — but insists it is *accessible*: the observer IS the underlying reality, and liberation (*moksha*) consists in recognizing this identity. Every Hindu and Buddhist soteriology rejects permanent inaccessibility.

Only Spinoza (whose fully determinate substance is naturally definite) and Worrall's epistemic structural realism (which posits real but unknowable natures behind structures) provide genuine philosophical support for Claim 4.

0.10.6 What's genuinely new

The paper's philosophical lineage is not a single line of descent but a mosaic. Its structural epistemology draws from Kant through Wittgenstein to Wolpert. Its complementarity thesis combines Bohr's physics with Cusa's coincidence of opposites and Daoist yin-yang. Its projection metaphysics reaches through Plato's cave and Plotinus's emanation to Advaita Vedanta's maya-Brahman distinction.

What is genuinely new is the *combination*: accepting Bohr's complementarity while insisting on Einstein's realism, grounding Kantian limits in physical structure while claiming knowledge of noumenal character, embracing Cusanian learned ignorance while grounding it in a concrete physical mechanism — the Gödel → Wolpert → causal partition chain described above. The traditions reveal that this combination creates a productive philosophical tension — the paper claims to know the character of what it proves unknowable. Whether this tension is a contradiction (as Hegel would insist), a residual Platonism (as Nietzsche would charge), or a legitimate achievement of learned ignorance (as Cusa would celebrate) may be the framework's deepest philosophical question.

0.11 Frequently Asked Questions

"Isn't this just another interpretation of quantum mechanics?" No. Interpretations (Copenhagen, Many-Worlds, Bohmian mechanics) accept the quantum formalism and disagree about what it *means*. This framework *derives* the quantum formalism from non-quantum

premises and proves the derivation is the only possible one. It makes quantitative predictions — the value of $\hbar$, dark energy evolution, and gravitational wave echoes — that interpretations do not, and it accounts for the dark sector's gravitational budget as an automatic consequence.

"Doesn't Bell's theorem rule out hidden variable theories?" Bell's theorem rules out *local* hidden variable theories satisfying a specific factorizability condition. This framework violates that factorizability — not through faster-than-light signals, but because P-indivisible joint dynamics don't permit the decomposition Bell's theorem assumes. The framework reproduces exactly Tsirelson's bound (the maximum quantum violation), no more and no less.

"If everything is deterministic underneath, where does randomness come from?" From ignorance. The total system is deterministic, but the observer can't access the hidden sector. Different hidden states compatible with the same visible state lead to different outcomes. The observer must assign probabilities — not because the universe is random, but because their information is incomplete.

"What about dark matter?" The framework doesn't set out to explain dark matter, but it produces a natural account of why ~95% of the universe's gravitational budget has no source in particle physics. The boundary entropy — the modes traced out to produce QM — has thermal energy equal to the critical density. Crucially, this thermal energy is computed from purely classical quantities (the number of boundary modes and the classical horizon temperature) — it exists before the trace-out, unlike the QFT zero-point energy, which is an artifact of the emergent description. The boundary entropy is therefore gravitationally active (it curves spacetime at the classical level) but invisible to the emergent QFT (the boundary modes were traced out to produce it). The baryonic sector accounts for ~5% of ρ_{crit}; the rest is boundary entropy. The uniform component is dark energy (Part II). Whether the matter-induced *redistribution* of this entropy produces local effects resembling dark matter at galaxy scales is an open question requiring a quantitative force law beyond the axioms. What the framework proves is that the total dark sector budget is an automatic consequence of the trace-out — not an additional input.

"Why does gravity 'see' the classical vacuum energy and not the quantum one?" Because the spacetime metric exists at the classical level, *before* the quantum description emerges. The quantum zero-point energy is a feature of the observer's compressed description. It's real for quantum experiments (the Casimir effect, the Lamb shift) but doesn't appear in the stress-energy tensor that governs curvature. The 10^{122} discrepancy is the information compression ratio — the entropy of the observer's blind spot.

"How can the paper claim reality is 'definite' if it's permanently inaccessible?" This is the paper's most philosophically contested thesis (see *Philosophical Lineage* above). The paper's defense is that the claim follows from the derivation's own logic: the axioms posit deterministic dynamics on a phase space, and the theorem shows that quantum indeterminacy arises from tracing out part of that phase space — not from any indeterminacy in the underlying evolution. The "definiteness" is a consequence of the starting premises, not a speculative addition. Whether those premises are the right ones to start from is, of course, an open question — but within the framework, Claim 4 is a theorem, not an assumption.

"Doesn't holographic physics show that spacetime comes from entanglement?" This is probably the strongest objection to the framework's ordering — classical spacetime first, quantum mechanics second. The Ryu-Takayanagi formula says entanglement entropy equals boundary area divided by $4G$. Van Raamsdonk argued that reducing entanglement disconnects spacetime. Programs like ER=EPR and "it from qubit" read these results as evidence that quantum entanglement is prior to geometry.

The framework offers an alternative reading. If the quantum description is produced by tracing out over a geometric boundary, then *of course* its entanglement entropy is proportional to the

boundary's area — the information content of the trace-out is set by the number of modes crossing the boundary, which scales with area. The Ryu-Takayanagi formula, on this account, isn't a hint that entanglement builds geometry; it's a consequence of the fact that geometry built the entanglement. The Bekenstein-Hawking entropy $S = A/(4\,l_p^2)$, which the paper derives, is exactly this statement.

The correlation between entanglement and geometry is real either way. The question is which direction the arrow of explanation points. The two orderings make different empirical predictions: if geometry emerges from entanglement, spacetime structure should break down at high energy; if quantum mechanics emerges from geometry, the quantum description should break down near the discreteness scale while the geometric substratum persists. The paper's gravitational wave echo prediction provides one test. And as discussed above, the cosmological constant dissolution is arguably existing evidence: the geometry-first ordering predicts the observed vacuum energy; the quantum-first ordering produces the 10^{122} discrepancy and has no solution.

0.12 What This Means

The search for a "theory of everything" that unifies quantum mechanics and general relativity has assumed that the two theories describe the same level of reality and must be reconciled there. This paper argues that assumption is wrong. The two theories operate at different levels — one fundamental, one emergent — and their apparent contradiction is the information-theoretic cost of being an observer trapped inside the system you're trying to describe.

The apparent conflict between QM and GR, the 10^{122} vacuum energy discrepancy, and the dark sector are three faces of the same fact. The first is the existence of the trace-out. The second is its information compression ratio. The third is its gravitational occlusion fraction. All three are mandatory consequences of observational incompleteness — and all three match what we observe.

The universe is not broken. We are observing it from within.

This is a companion overview to "The Incompleteness of Observation: Why Quantum Mechanics and General Relativity Cannot Be Unified From Within" (Maybaum, March 2026), which presents the formal arguments with detailed derivations. The philosophical lineage section draws on a systematic analysis mapping the paper's claims against the major traditions in Western, Eastern, and contemporary philosophy of science.

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